Lesson 11.2 - How to Find Cohen's D to Determine the Effect Size

> **Video:** How to find Cohen's D to determine the Effect Size Using Pooled Standard Deviation by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: is the difference big, not just real?

Lesson 11.1 asked **whether** two means differ. This lesson asks **how much** they differ, and why that is a separate question.

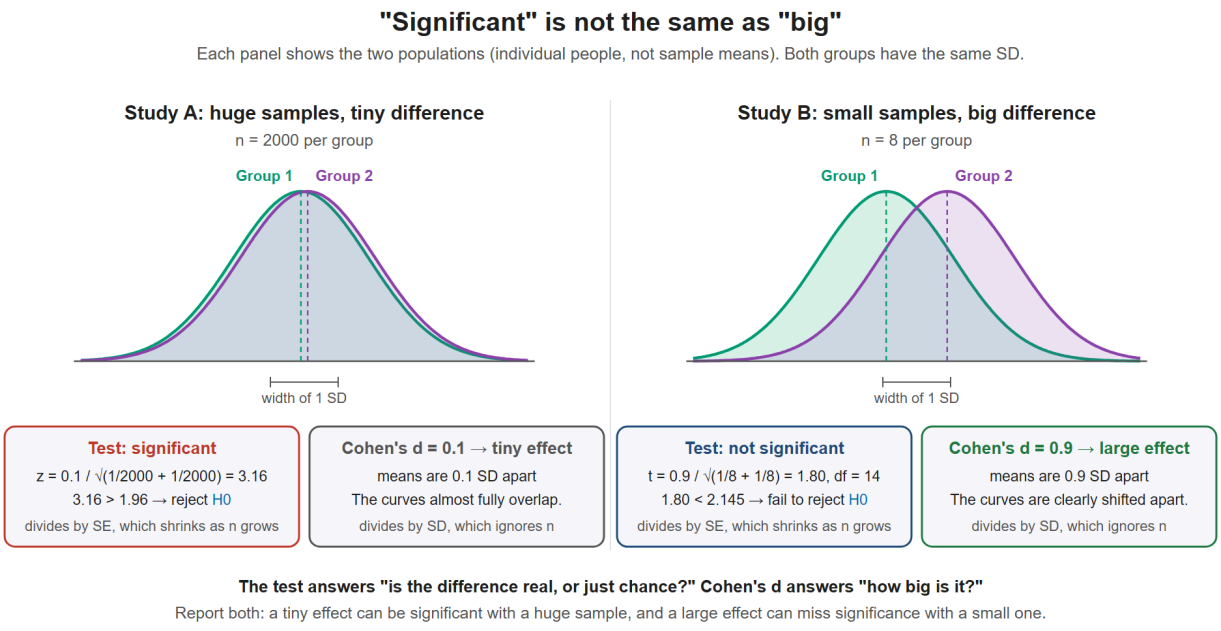
1. A test answers "real or luck?", not "big or small?". Rejecting H_0 means the gap is too large to blame on chance. It doesn't say the gap is large enough to matter.

2. Significance depends on sample size. Lesson 10.4 showed that the same gap gives a bigger test statistic with a bigger sample, because the standard error shrinks like $1/\sqrt{n}$. So:

- With **huge** samples, even a tiny, unimportant difference becomes "significant".
- With **small** samples, even a large, important difference can fail to reach significance.

3. Cohen's d removes the sample size. It divides by the **standard deviation** (how much individual people vary), not the **standard error** (how much sample means vary):

	Test statistic (Lesson 11.1)	Cohen's d (this lesson)
Divides the gap by	standard **error**, $\approx s\sqrt{1/n_1 + 1/n_2}$	standard **deviation**, s_{pooled}
Gets bigger with more data?	**Yes**, because the SE shrinks	**No**, because the SD stays about the same
Answers	Is there a difference?	How big is the difference?



Study A has a tiny effect ($d = 0.1$), but with 2000 people per group it is still significant. Study B has a large effect ($d = 0.9$), but with only 8 people per group it misses significance. The test and Cohen's d answer different questions.

4. Why "standard deviation units"? Raw gaps depend on the units: 3 points on a test, 4 pounds of weight. Dividing by the SD makes d **unitless**, so you can compare effects across different studies and measurements. $d = 0.5$ means the same thing everywhere: the averages are half a standard deviation apart.

5. Why pool the standard deviations? d needs **one** SD to divide by, but each group has its own s. Assuming both groups vary about the same amount, we combine them into a single best estimate, giving the bigger group more weight.

> **Report both.** A full answer says whether the difference is real (test or p-value) **and** how big it is (Cohen's d). For example: "significant at the 5% level, with a large effect, $d = 0.85$."

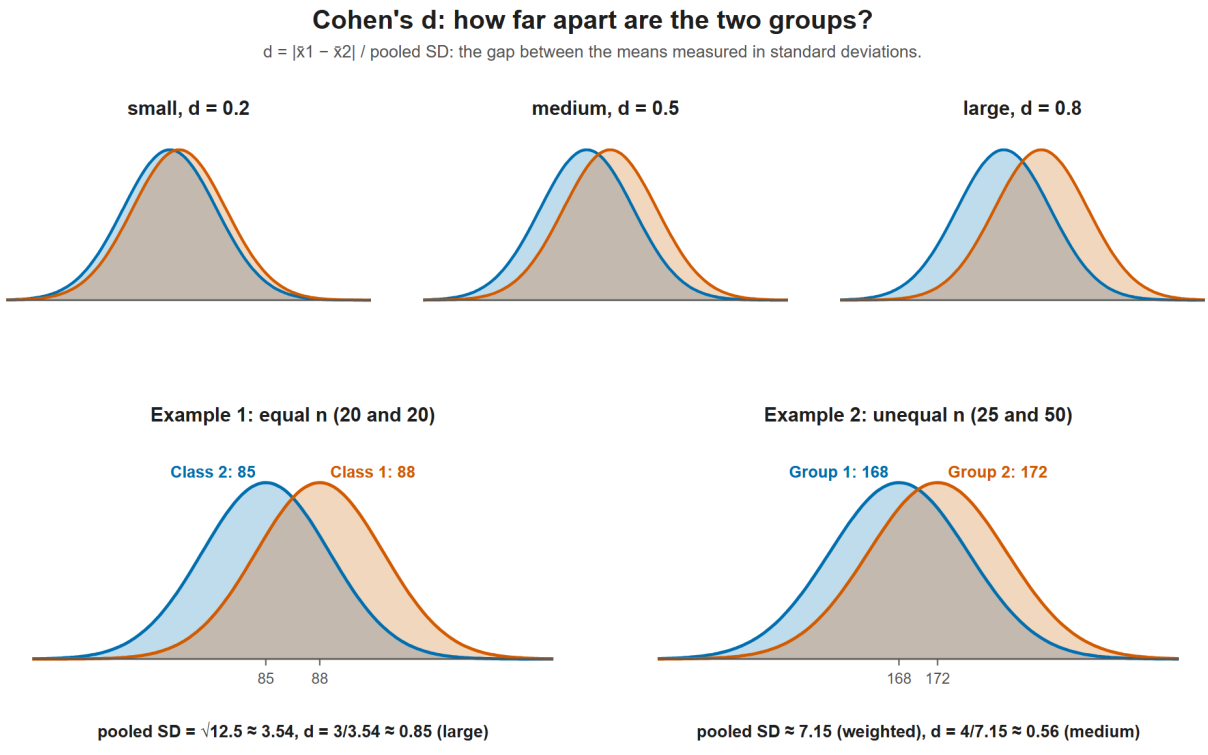
What Cohen's d measures

Cohen's d expresses the difference between two means **in standard-deviation units**:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}}$$

- $d = 1$: the means are 1 standard deviation apart.
- $d = 2$: 2 standard deviations apart.
- $d = 0.5$: half a standard deviation apart.

d	Effect size
≈ 0.2	Small
≈ 0.5	Medium
≥ 0.8	Large



Cohen's d measures how far apart the two curves sit, in units of the pooled standard deviation. The more the curves overlap, the smaller the effect. Example 1 (d ≈ 0.85) is a large effect; Example 2 (d ≈ 0.56) is medium.

Pooled standard deviation

General formula (use it when the sample sizes differ):

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Shortcut (only when $n_1 = n_2$, or nearly so):

$$s_{\text{pooled}} = \sqrt{\frac{s_1^2 + s_2^2}{2}}$$

Example 1 - two classes, equal sample sizes

	Class 1	Class 2
n	20	20
$\bar{x}$	88	85
s	3	4

Pooled SD (general formula):

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{19(3^2) + 19(4^2)}{20 + 20 - 2}} = \sqrt{\frac{171 + 304}{38}} = \sqrt{\frac{475}{38}} = \sqrt{12.5} \approx 3.5355$$

Shortcut (equal n): $\sqrt{\frac{s_1^2 + s_2^2}{2}} = \sqrt{\frac{9 + 16}{2}} = \sqrt{12.5} \approx 3.5355$, the same answer ✓

Cohen's d:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}} = \frac{88 - 85}{3.5355} \approx 0.85 \;\Rightarrow\; \textbf{large effect}$$

Example 2 - two groups, unequal sample sizes

	Group 1	Group 2
n	25	50
$\bar{x}$ (lb)	168	172
s (lb)	5	8

Pooled SD (must use the general formula, because the sizes differ):

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{24(5^2) + 49(8^2)}{25 + 50 - 2}} = \sqrt{\frac{600 + 3136}{73}} = \sqrt{\frac{3736}{73}} \approx 7.154$$

The shortcut would give $\sqrt{(25 + 64)/2} \approx 6.671$, which is **wrong** here. Group 2 is twice as large, so its SD (8) should carry more weight, and the correct pooled SD (7.15) is closer to 8.

Cohen's d:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}} = \frac{|168 - 172|}{7.154} = \frac{4}{7.154} \approx 0.56 \;\Rightarrow\; \textbf{medium effect}$$

Key takeaways

- d = |difference in means| ÷ pooled SD, a **unitless** measure of how big the difference is.
- Benchmarks: 0.2 small, 0.5 medium, 0.8+ large.
- Use the weighted pooled-SD formula unless the sample sizes are equal.
- A hypothesis test asks *whether* there is a difference. Cohen's d asks *how large* it is.
- The test divides by the SE, which shrinks as n grows. d divides by the SD, which doesn't. So "significant" doesn't mean "big", and "not significant" doesn't mean "small".